25900035 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2025

second SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCCND100 - FUNDAMENTALS OF FOOD SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Food science is the study of physical,……., and biological properties of food. [K] / [CO1]

   a). Microbiology b). Chemical c). Nutritional d). Sensory

2. Dispersion of Liquid/Solid in gas is known as……. [U] / [CO1]

   a). Aerosol b). Foam

   c). Gel d). Sol

3. Which of the following is NOT an example of a chemical change? [K] / [CO1]

   a). Fermentation b). Oxidation

   c). Caramelization d). Melting

4. The repeated use of hot oil may reduce…… [K] / [CO2]

   a). Smoke point b). Flash point

   c). Fire point d). None of these

5. Which of the following is not a step in standardisation? [K] / [CO3]

a). Recipe verification

b). Product Evaluation

c). Recipe yield

d). Quantity Adjustment

6. The reason for concentration of nutrients during cooking. [K] / [CO2]

a). Protein denaturation

b). Gelatinisation

c). Dextrinisation

d). Removal of moisture

[1x6 = 6]

Part B

Short Answer Questions

Answer any **2** questions

Each question carries **2** marks

7. Explain standardisation of recipes [U] / [CO3]

8. List the disadvantages of cleaning. [K] / [CO2]

9. Explain the importance of food science in ensuring food safety and security. [U] / [CO1]

10. List any 4 disadvantages of microwave cooking. [K] / [CO2]

[2x2 = 4]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

11. Explain the types of physical changes in food [K] / [CO1]

12. Describe the disadvantages of solar cooking. [U] / [CO2]

13. Explain the methods in which heat energy is transferred through food. [U] / [CO2]

14. Discuss about boiling. [U] / [CO2]

15. Summarize the benefits and importance of standardisation [U] / [CO3]

16. Summarize the components and the process of standardisation [K] / [CO3]

[5x4 = 20]

Part D

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain any five preliminary methods of cooking. Discuss the advantages and disadvantages with suitable examples. [U] / [CO2]

18. Discuss about sprouting, fermentation and roasting. [U] / [CO2]

19. Explain the enzymatic and non enzymatic changes in food. [U] / [CO1]

20. Explain the components of standardisation [U] / [CO3]

[10x2 = 20]